Declarative vs. Procedural Knowledge

Prompt and Completion:
The order of operations or PEMDAS is
Parentheses Exponents Multiplication Division Addition Subtraction

Prompt and Completion:
(1 + 1) × 2 = **3**

Figure 7: GPT-3's completion for two prompts testing knowledge of the order of operations. The blue underlined bold text is the autocompleted response from GPT-3. While it *knows about* the order of operations, it sometimes does not *know how* to apply its knowledge.

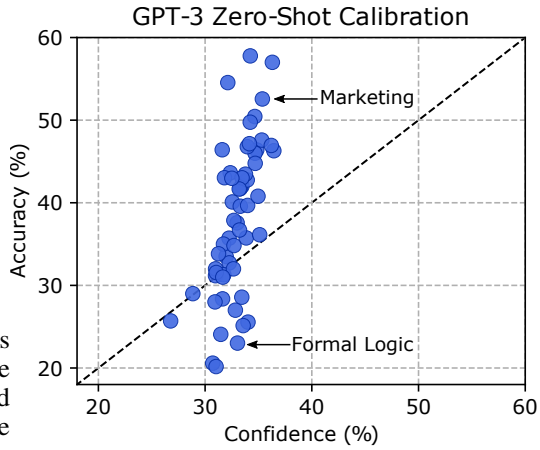

Figure 8: GPT-3's confidence is a poor estimator of its accuracy and can be off by up to 24%.

lowest-accuracy tasks are STEM subjects that emphasize mathematics or calculations. We speculate that is in part because GPT-3 acquires declarative knowledge more readily than procedural knowledge. For example, many questions in Elementary Mathematics require applying the order of operations for arithmetic, which is described by the acronym PEMDAS (Parentheses Exponents Multiplication Division Addition Subtraction). In Figure 7, we confirm that GPT-3 is *aware* of the acronym PEMDAS. However, it does not consistently *apply* PEMDAS to actual problems. On the other hand, procedural understanding is not its only weak point. We find that some verbal tasks such as Moral Scenarios from Hendrycks et al. (2020) and Professional Law also have especially low accuracy.

Our test also shows that GPT-3 acquires knowledge quite unlike humans. For example, GPT-3 learns about topics in a pedagogically unusual order. GPT-3 does better on College Medicine (47.4%) and College Mathematics (35.0%) than calculation-heavy Elementary Mathematics (29.9%). GPT-3 demonstrates unusual breadth, but it does not master a single subject. Meanhwhile we suspect humans have mastery in several subjects but not as much breadth. In this way, our test shows that GPT-3 has many knowledge blindspots and has capabilities that are lopsided.

Calibration. We should not trust a model's prediction unless the model is calibrated, meaning that its confidence is a good estimate of the actual probability the prediction is correct. However, large neural networks are often miscalibrated (Guo et al., 2017), especially under distribution shift (Ovadia et al., 2019). We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject. We show the results in Figure 8, which demonstrates that GPT-3 is uncalibrated. In fact, its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects. Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%. Models are only somewhat more calibrated in the few-shot setting, as shown in Appendix A. These results suggest that model calibration has wide room for improvement.

5 Discussion

Multimodal Understanding. While text is capable of conveying an enormous number of concepts about the world, many important concepts are conveyed mainly through other modalities, such as images, audio, and physical interaction (Bisk et al., 2020). Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format. However, as models gain the ability to process multimodal inputs, benchmarks should be designed to reflect this change. One such benchmark could be a "Turk Test," consisting of Amazon Mechanical Turk Human Intelligence Tasks. These are well-defined tasks that require models to interact with flexible formats and demonstrate multimodal understanding.

The Internet as a Training Set. A major distinction between our benchmark and previous multitask NLP benchmarks is that we do not require large training sets. Instead, we assume that models have acquired the requisite knowledge from reading vast quantities of diverse text from the Internet. This

process is typically called pretraining, but it can be thought of as training in its own right, where the downstream evaluation is demonstrating whatever knowledge we would expect a human to pick up from reading the same text.

This motivates us to propose a methodological change so that models are trained more like how humans learn. While most previous machine learning benchmarks have models learn from a large question bank, humans primarily learn new subjects by reading books and listening to others talk about the topic. For specialized subjects such as Professional Law, massive legal corpora are available, such as the 164-volume legal encyclopedia *Corpus Juris Secundum*, but there are fewer than 5,000 multistate bar exam questions available. Learning the entire law exclusively through a small number of practice tests is implausible, so future models must learn more during pretraining.

For this reason we assess pretrained models in a zero-shot, few-shot, or transfer setting and we provide a dev, val, and test set for each task. The dev set is used for few-shot prompts, the val set could be used for hyperparameter tuning, and the test set is used to compute the final accuracy. Importantly, the format of our evaluation is not identical to the format in which information is acquired during pretraining. This has the benefit of obviating concerns about spurious training set annotation artifacts (Geirhos et al., 2020; Hendrycks et al., 2019b) and is in stark contrast to the previous paradigm of identically distributed training and test sets. This change also enables collecting a much more extensive and diverse set of tasks for evaluation. We anticipate our methodology becoming more widespread as models improve at extracting information from diverse online sources.

Model Limitations. We find that current large-scale Transformers have wide room for improvement. They are notably poor at modeling human (dis)approval, as evident by the low performance on the Professional Law and Moral Scenarios tasks. For future systems to be aligned with human values, high performance on these tasks is crucial (Hendrycks et al., 2020), so future research should especially aim to increase accuracy on these tasks. Models also have difficulty performing calculations, so much so that they exhibit poor performance on Elementary Mathematics and many other STEM subjects with "plug and chug" problems. Additionally, they do not match expert-level performance (90%) on any subject, so for all subjects it is subhuman. On average, models are only now starting to move beyond random-chance accuracy levels.

Addressing these shortcomings may be challenging. To illustrate this, we attempted to create a better Professional Law model by pretraining on specialized data but achieved only limited success. We collected approximately 2,000 additional Professional Law training examples. After fine-tuning a RoBERTa-base model (Liu et al., 2019) using this custom training set, our model attained 32.8% test accuracy. To test the impact of additional specialized training data, we also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus `case.law`, but after fine-tuning it only attained 36.1% accuracy. This suggests that while additional pretraining on relevant high quality text can help, it may not be enough to substantially increase the performance of current models.

It is unclear whether simply scaling up existing language models will solve the test. Current understanding indicates that a $10\times$ increase in model size must be accompanied by an approximate $5\times$ increase in data (Kaplan et al., 2020). Aside from the tremendous expense in creating multi-trillion parameter language models, data may also become a bottleneck, as there is far less written about esoteric branches of knowledge than about everyday situations.

6 Conclusion

We introduced a new test that measures how well text models can learn and apply knowledge encountered during pretraining. By covering 57 subjects at varying levels of difficulty, the test assesses language understanding in greater breadth and depth than previous benchmarks. We found that it has recently become possible for models to make meaningful progress on the test, but that state-of-the-art models have lopsided performance and rarely excel at any individual task. We also showed that current models are uncalibrated and have difficulty with tasks that require calculations. Worryingly, models also perform especially poorly on socially relevant subjects including morality and law. Our expansive test can help researchers pinpoint important shortcomings of models, making it easier to gain a clearer picture of state-of-the-art capabilities.

ACKNOWLEDGEMENTS

We would like to thank the following for their helpful comments: Oyvind Tafjord, Jan Leike, David Krueger, Alex Tamkin, Girish Sastry, and Henry Zhu. DH is supported by the NSF GRFP Fellowship and an Open Philanthropy Project Fellowship. This research was also supported by the NSF Frontier Award 1804794.

REFERENCES

M. G. Bellemare, Y. Naddaf, J. Veness, and M. Bowling. The arcade learning environment: An evaluation platform for general agents (extended abstract). *J. Artif. Intell. Res.*, 47:253–279, 2013.

Y. Bisk, R. Zellers, R. L. Bras, J. Gao, and Y. Choi. Piqa: Reasoning about physical commonsense in natural language, 2019.

Y. Bisk, A. Holtzman, J. Thomason, J. Andreas, Y. Bengio, J. Chai, M. Lapata, A. Lazaridou, J. May, A. Nisnevich, N. Pinto, and J. Turian. Experience grounds language, 2020.

T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei. Language models are few-shot learners, 2020.

P. Clark, I. Cowhey, O. Etzioni, T. Khot, A. Sabharwal, C. Schoenick, and O. Tafjord. Think you have solved question answering? try arc, the ai2 reasoning challenge. *ArXiv*, abs/1803.05457, 2018.

P. Clark, O. Etzioni, D. Khashabi, T. Khot, B. D. Mishra, K. Richardson, A. Sabharwal, C. Schoenick, O. Tafjord, N. Tandon, S. Bhakthavatsalam, D. Groeneveld, M. Guerquin, and M. Schmitz. From 'f' to 'a' on the n.y. regents science exams: An overview of the aristo project. *ArXiv*, abs/1909.01958, 2019.

J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *ArXiv*, abs/1810.04805, 2019.

R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann. Shortcut learning in deep neural networks, 2020.

C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. *ICML*, 2017.

D. Hendrycks, M. Mazeika, and T. Dietterich. Deep anomaly detection with outlier exposure. *ICLR*, 2019a.

D. Hendrycks, K. Zhao, S. Basart, J. Steinhardt, and D. Song. Natural adversarial examples. *ArXiv*, abs/1907.07174, 2019b.

D. Hendrycks, C. Burns, S. Basart, A. Critch, J. Li, D. Song, and J. Steinhardt. Aligning ai with shared human values, 2020.

L. Huang, R. L. Bras, C. Bhagavatula, and Y. Choi. Cosmos qa: Machine reading comprehension with contextual commonsense reasoning, 2019.

J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei. Scaling laws for neural language models, 2020.

D. Khashabi, T. Khot, A. Sabharwal, O. Tafjord, P. Clark, and H. Hajishirzi. Unifiedqa: Crossing format boundaries with a single qa system, 2020.

T. Khot, P. Clark, M. Guerquin, P. Jansen, and A. Sabharwal. Qasc: A dataset for question answering via sentence composition, 2019.

A. Kumar, P. Liang, and T. Ma. Verified uncertainty calibration, 2019.